Recap from last class

March 25, 2026

Theorem (Extreme Value Theorem):
If $f: X \subseteq \mathbb{R}^n \to \mathbb{R}$ is continuous and X is compact, then f attains a global max and a global min on X.
(i.e. there exist $\vec{a}$ and $\vec{b}$ in X such that for all $\vec{x}$ in X
global min $\to$ $f(\vec{a}) \leq f(\vec{x}) \leq f(\vec{b})$ $\leftarrow$ global max)
Moreover, all global extrema must occur either at a boundary point, or at the site of a local extremum (i.e. at a critical point of f).

e.g. Find the global extrema of $f(x) = x^3 - 3x^2$ on $[-2, 2]$

1) Find boundary points of X
2) Find critical points of f inside X
3) Evaluate f on each boundary point and critical point
4) The largest value of f from 3) is the global max, and the smallest value of f from 3) is the global min

1) $\operatorname{bd}(X) = \{-2, 2\}$
2) f is C^1 (continuous derivatives) $f'(x) = 3x^2 - 6x$
$f'(x) = 0 \Rightarrow 0 = 3x(x-2)$ $x = 0,\ x = 2$

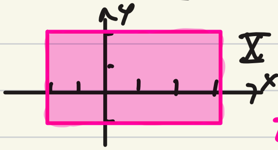
3) $f(-2) = (-2)^3 - 3(-2)^2 = -8 - 12 = -20$
$f(2) = (2)^3 - 3(2)^2 = 8 - 12 = -4$
$f(0) = 0$

4) $f(-2) = -20$ is the global min of f over X

$f(0) = 0$ is the global max of f over X

Example from last class continued

e.g. Find the global extrema of $f(x,y) = (x-1)^2 + y^2$
over $X = \{(x,y) \in \mathbb{R}^2 : -2 \le x \le 3 \ , \ -1 \le y \le 2\}$



1) $\mathrm{bd}(X) = ?$

$\{(3,y) \in \mathbb{R}^2 : -1 \le y \le 2\} \cup \{(x,2) \in \mathbb{R}^2 : -2 \le x \le 3\}$
$\cup \{(-2,y) \in \mathbb{R}^2 : -1 \le y \le 2\} \cup \{(x,-1) \in \mathbb{R}^2 : -2 \le x \le 3\}$

2) Critical points: f is continuously differentiable C^1.

$\vec{\nabla} f = (2(x-1), \ 2y)$

$\vec{\nabla} f = 0 \Rightarrow \ 2(x-1) = 0, \ x = 1$ and $2y = 0, \ y = 0$

critical points: $\{(1,0)\}$ only one critical point

* Important check here: the critical point is in X

* 3) $f(1,0) = 0$

3) Find values of f at boundary points and critical points:

at critical points: at $(1,0)$ $f(1,0) = 0$

at boundary:

over $\{(3,y) \in \mathbb{R}^2 : -1 \le y \le 2\}$: $f(3,y) = (3-1)^2 + y^2$
$= 4 + y^2 \quad -1 \le y \le 2$

* 4) max and min of $4 + y^2$ over $[-1,2]$:
min at vertex $4 + (0)^2 = 4$
max is one of $4 + (-1)^2 = 5$ or $4 + (2)^2 = 8$

over $\{(x,2) \in \mathbb{R}^2 : -2 \le x \le 3\}$: $f(x,2) = (x-1)^2 + 4 \quad -2 \le x \le 3$

* 4) max of $(x-1)^2 + 4$ over $-2 \le x \le 3$ is $f(-2,2) = 13$
min of $(x-1)^2 + 4$ over $-2 \le x \le 3$ is $f(1,2) = 4$

over $\{(-2,y) \in \mathbb{R}^2 : -1 \le y \le 2\}$: $f(-2,y) = 9 + y^2 \quad -1 \le y \le 2$

* 4) max of $9 + y^2$ over $-1 \le y \le 2$ is $f(-2,2) = 13$
min of $9 + y^2$ over $-1 \le y \le 2$ is $f(-2,0) = 9$

over $\{(x,-1) \in \mathbb{R}^2 : -2 \le x \le 3\}$: $f(x,-1) = (x-1)^2 + 1 \quad -2 \le x \le 3$

* 4) max of $(x-1)^2 + 1$ over $-2 \le x \le 3$ is $f(-2,-1) = 10$
min of $(x-1)^2 + 1$ over $-2 \le x \le 3$ is $f(1,-1) = 1$

4) largest and smallest values of f global max is 13 global min is 0

Lagrange Multipliers

e.g. Find the extrema of $f(x,y) = x + y$ over the unit disc
$\Sigma = \{(x,y) : x^2+y^2 \leq 1\}$

Σ is compact, so we can use the methods from last class

1) $bd(\Sigma) = \{(x,y) : x^2+y^2=1\}$

2) Critical points : f is C^1 (continuously differentiable), so $\vec{\nabla} f$ is defined everywhere, and we only need to look for $\vec{\nabla} f = \vec{0}$.
$\vec{\nabla} f = (1,1)$ so no critical point

3) Find values of f at critical points and boundary points :
Need to find all values of $f(x,y)$ for $x^2+y^2=1$
Parametrize the circle $(x,y) = (\cos\theta, \sin\theta)$ for $\theta \in [0,2\pi]$
instead of finding all values of $f(\cos\theta, \sin\theta) = \cos\theta + \sin\theta$, we only need to find the max and min over $0 \leq \theta \leq 2\pi$:

1*) $bd([0,2\pi]) = \{0, 2\pi\}$

2*) Critical points : $\frac{d}{d\theta}(\cos\theta + \sin\theta) = -\sin\theta + \cos\theta = 0$
$\Rightarrow \sin\theta = \cos\theta \Rightarrow \theta = \pi/4,\ 5\pi/4$

3*) Find values of f at $\theta = 0, 2\pi, \pi/4$ and $5\pi/4$

$f(\cos 0, \sin 0) = f(\cos 2\pi, \sin 2\pi) = 1$

$f(\cos\frac{\pi}{4}, \sin\frac{\pi}{4}) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$

$f(\cos\frac{5\pi}{4}, \sin\frac{5\pi}{4}) = \frac{-1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\sqrt{2}$

4*) Global max and min of $f(\cos\theta, \sin\theta)$ over $0 \leq \theta \leq 2\pi$:
global max is $f(\cos\frac{\pi}{4}, \sin\frac{\pi}{4}) = \sqrt{2}$
global min is $f(\cos\frac{5\pi}{4}, \sin\frac{5\pi}{4}) = -\sqrt{2}$

4) Global max and min of $f(x,y)$ over Σ :
global max is $f(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) = \sqrt{2}$ global min is $f(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}) = -\sqrt{2}$

Imagine if now we look for the global max and min of $f(x,y,z) = 2x + y - z$ over the closed unit ball

$$\mathbb{X} = \{(x,y,z) : x^2+y^2+z^2 \le 1\}.$$

$\vec{\nabla} f = (2,1,-1)$ is never $\vec{0}$, so we "only" need to evaluate f over the boundary of $\mathbb{X}$, $bd(\mathbb{X}) = \{(x,y,z) : x^2+y^2+z^2=1\}$

We can use polar coordinates in 3-D to parametrize the sphere $bd(\mathbb{X})$ by

$$x = \cos\theta\cos\varphi \qquad y = \cos\theta\sin\varphi \qquad z = \sin\theta$$

over $\{(\theta,\varphi) : -\pi/2 \le \theta \le \pi/2 \ , \ 0 \le \varphi \le 2\pi\}$

$\theta \leftrightarrow$ lattitude , $\varphi \leftrightarrow$ longitude : (in terms of Earth coordinates)

There is another (usually simpler) method to solve this type of problems

def: <u>Constrained optimization</u> is the process of maximizing and/or minimizing an <u>objective function</u> $f(\vec{x}) : \mathbb{X} \subseteq \mathbb{R}^n \to \mathbb{R}$ while imposing the condition that $\vec{x}$ must satisfy a constraint equation $g(\vec{x}) = c$ (for some function $g : \mathbb{X} \subseteq \mathbb{R}^n \to \mathbb{R}$ and a constant $c \in \mathbb{R}$).

We could also have several constraints: $g_1(\vec{x}) = c_1, \dots, g_k(\vec{x}) = c_k$.

e.g. on $\mathbb{X} = \{(x,y) : x^2+y^2 \le 1\}$ we want to find global max and min for $f(x,y) = x+y$ subject to the constraint $g(x,y) = x^2+y^2 = 1$

e.g. on $\mathbb{X} = \mathbb{R}^3$ we want to optimize the function $f(x,y,z) = 2x+y-z$ subject to the constraint $g(x,y,z) = x^2+y^2+z^2 = 1$

e.g. Which point is closest to the origin and which is farthest when you intersect the ellipsoid $\frac{x^2}{4} + \frac{y^2}{9} + z^2 = 1$ with the parabolloid $z = x^2+y^2$

$\mathbb{X} = \mathbb{R}^3 \qquad f(\vec{x}) = \|\vec{x}\| = \sqrt{x^2+y^2+z^2} \qquad g(\vec{x}) = \ ?$

$g_1(\vec{x}) = \frac{x^2}{4} + \frac{y^2}{9} + z^2$

we might as well optimize $f(\vec{x}) = x^2+y^2+z^2$

$g_2(\vec{x}) = x^2+y^2-z$

$g_1(\vec{x}) = 1 \qquad g_2(\vec{x}) = 0$

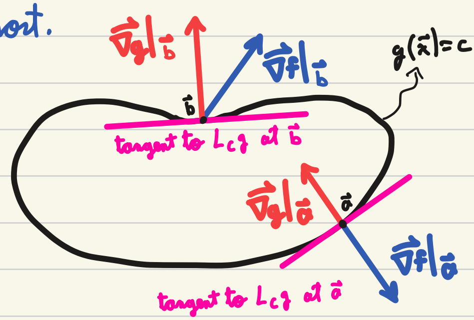
Theorem (Lagrange Multipliers): Suppose $X \subseteq \mathbb{R}^n$ is open and $f, g : X \to \mathbb{R}$ are functions of class C^1 (all first order partial derivatives exist and are continuous). If all of the following hold:

1) $\vec{a} \in L_c g = \{\vec{x} \in X : g(\vec{x}) = c\}$ (level set of g at height c),
2) $\vec{\nabla} g(a) \neq \vec{0}$ and
3) f restricted to $L_c g$ has a local extremum at $\vec{a}$,

then there is some $\lambda \in \mathbb{R}$ such that $\vec{\nabla} f|_{\vec{a}} = \lambda \vec{\nabla} g|_{\vec{a}}$.

geometric idea of the theorem:

Recall that the gradient vector $\vec{\nabla} f$ points in the direction in which f increases the most.
So "any small displacement" $d\vec{x}$ that is parallel to $\vec{\nabla} f$ will produce a change in f.
More generally, "any small displacement" $d\vec{x}$ that is not perpendicular to $\vec{\nabla} f$ will change f.
On the other hand, $\vec{\nabla} g$ (when it is not 0) is a normal vector to the level set $g(\vec{x}) = c$,
so "any small displacement" $d\vec{x}$ along the level set will be perpendicular to $\vec{\nabla} g$



def: In a constrained optimization problem with objective function $f : X \to \mathbb{R}$ and restraint $g(\vec{x}) = c$, we say $\vec{a} \in X$ is a <u>constrained critical point</u> if

1) $g(\vec{a}) = c$,
2) $\vec{\nabla} g|_{\vec{a}} \neq \vec{0}$ and
3) there is some $\lambda \in \mathbb{R}$ such that $\vec{\nabla} f|_{\vec{a}} = \lambda \vec{\nabla} g|_{\vec{a}}$

Theorem: If $\vec{a} \in X$ is a local max or min of the function $f : X \subseteq \mathbb{R}^n \to \mathbb{R}$ subject to the constraint $g(\vec{x}) = c$, then $g(\vec{a}) = c$ and either

1) $\vec{a}$ is a <u>constrained critical point</u>, 2) $\vec{\nabla} g|_{\vec{a}} = \vec{0}$,
3) f is not differentiable at $\vec{a}$ or 4) g is not differentiable at $\vec{a}$

def: In a contrained optimization problem with objective function $f: X \to \mathbb{R}$ and restraint $g(\vec{x}) = c$, we say $\vec{a} \in X$ is a <u>contrained critical point</u> if 1) $g(\vec{a}) = c$,

2) $\vec{\nabla} g|_{\vec{a}} \neq \vec{0}$ and

3) there is some $\lambda \in \mathbb{R}$ such that $\vec{\nabla} f|_{\vec{a}} = \lambda \vec{\nabla} g|_{\vec{a}}$

Theorem: If $\vec{a} \in X$ is a local max or min of the function $f: X \subseteq \mathbb{R}^n \to \mathbb{R}$ subject to the contraint $g(\vec{x}) = c$, then $g(\vec{a}) = c$ and either

1) $\vec{a}$ is a <u>contrained critical point</u>, 2) $\vec{\nabla} g|_{\vec{a}} = \vec{0}$,

3)* f is not differentiable at $\vec{a}$ or 4)* g is not differentiable at $\vec{a}$

e.g. Optimize $x+y$ over the unit circle

Objective function is $f(x,y) = x + y$ domain X can be all $\mathbb{R}^2$

contraint $g(x,y) = x^2 + y^2 = 1$

* We don't need 3) or 4) since f and g are C^1 functions

1) Let's find all contrained critical points:

$\vec{\nabla} g = (2x, 2y)$, $\vec{\nabla} f = (1,1)$ and we need solutions to $\lambda(2x, 2y) = (1,1)$ so $2x\lambda = 1$ and $x=y$

$x = y = 1/(2\lambda)$

We also need to make sure to satisfy the contraint

so we need (x,y) such that $x = y = 1/(2\lambda)$ and $x^2+y^2=1$

There are only two solutions

$(x,y) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ (and $\lambda = 1/\sqrt{2}$) or $(x,y) = \left(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$ ($\lambda = -1/\sqrt{2}$)

2) Find $\vec{a} \in L_1 g$ such that $\vec{\nabla} g|_{\vec{a}} = \vec{0}$

$\vec{\nabla} g = \vec{0}$ only at $(0,0)$ which does not satisfy $x^2+y^2=1$

Conclusion: $f\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \sqrt{2}$ and $f\left(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right) = -\sqrt{2}$

so the max is $\sqrt{2}$ and the min is $-\sqrt{2}$

e.g. now we look for the global max and min of $f(x,y,z) = 2x + y - z$ over the closed unit ball $X = \{(x,y,z) : x^2+y^2+z^2 \leq 1\}$.

1) $bd(X) = \{(x,y,z) : x^2+y^2+z^2=1\}$

2) Critical points of f inside X:

$\vec{\nabla} f = (2, 1, -1)$ so no critical points.

3) Evaluate f at critical points and boundary:

We only care about the boundary, and what we want is to optimize $f(x,y,z)=2x+y-z$ subject to $g(x,y,z) = x^2+y^2+z^2 = 1$

Let's use Lagrange multipliers:

We want points (x,y,z) where

$$\vec{\nabla} f = (2,1,-1) = \lambda \vec{\nabla} g = \lambda(2x, 2y, 2z)$$

TBC ...